



CMM 2025 Integration Bee Qualification Test

Name:

Team:

You will have 10 minutes to complete the following 14 integrals. Since all of the following integrals are definite, please box your answer next to the corresponding integral so the graders can clearly delineate your answers, which must all be in the **simplest possible form** per the test-taking conventions. You may use the front and back of this page for scratch work as well as additional pieces of paper, but **do not** perform calculations next to the integral. Do not complete this test unless you intend to compete in the finals. **No computers, calculators, nor outside resources are permitted.**

Problem 1. $\int_0^{\infty} x^7 e^{-x^2} dx$

Problem 2. $\int_0^{\pi} \sin^5(x) dx$

Problem 3. $\int_{-3}^3 \frac{x^4}{1+e^{-x}} dx$

Problem 4. $\int_0^1 (\sin(\log x) + \cos(\log x)) dx$

Problem 5. $\int_{-1}^1 \frac{\cos(x)}{\arctan(x)} dx$

Problem 6. $\int_{\frac{\pi}{2}}^{\pi} \sqrt{1+\cos(x)} dx$

Problem 7. $\int_{-\infty}^0 e^{-\sqrt[6]{-x}} dx$

Problem 8. $\int_1^{\infty} \frac{1}{2025^{\lfloor x \rfloor}} dx$

Problem 9. $\int_4^5 \frac{2x+1}{(x-1)(x-2)(x-3)} dx$

Problem 10. $\int_0^1 x e^{x^2+e^{x^2}} dx$

Problem 11. $\int_0^{\frac{3}{2}} \lfloor x + \lfloor 2x \rfloor \rfloor dx$

Problem 12. $\int_{2025}^{2024} \ln(\{x\}) dx$

Problem 13. $\int_{98}^{99} x^{-\frac{2}{\log x}} dx$

Problem 14. $\int_{-\infty}^{\infty} e^{-x^2+3x} dx$